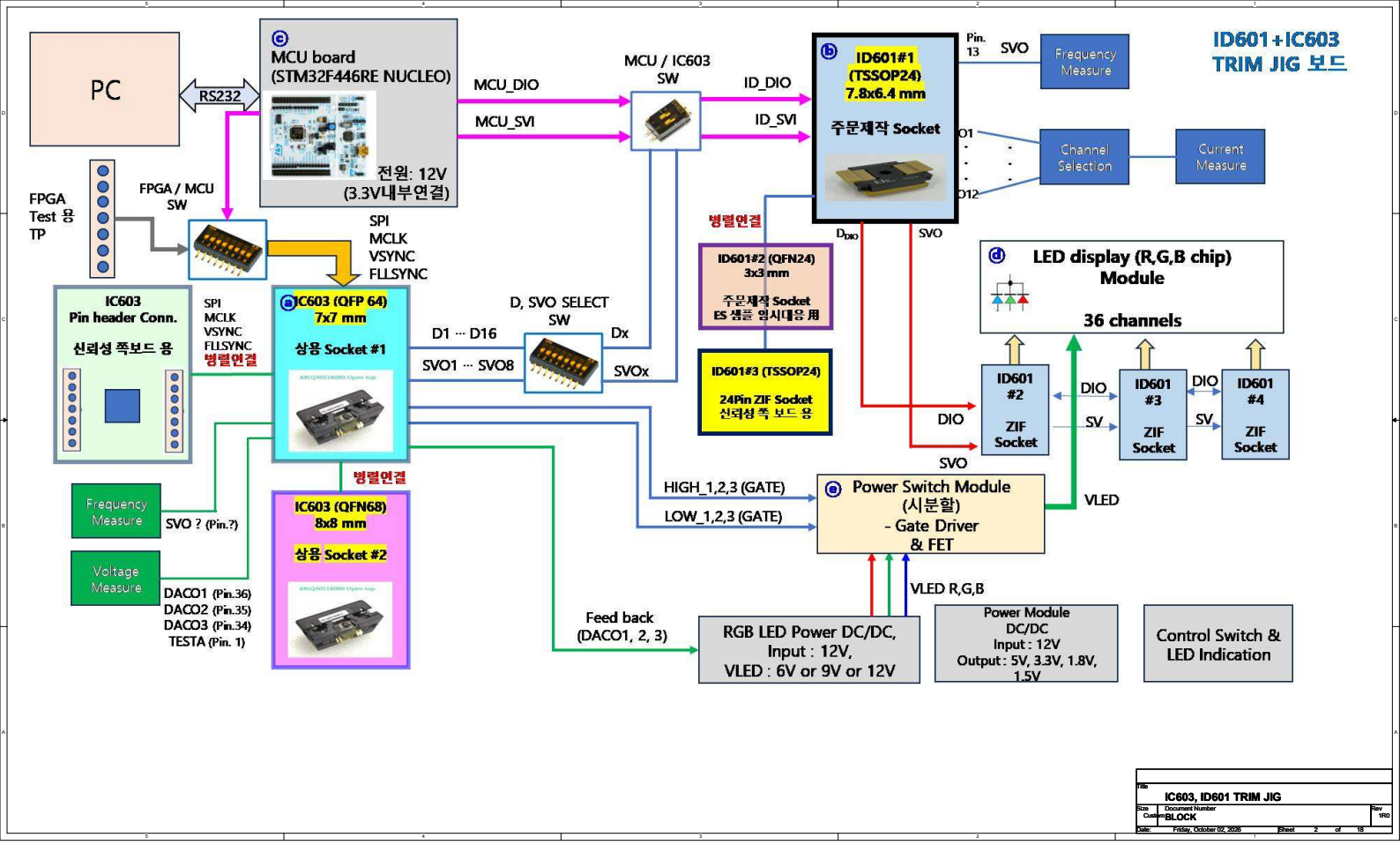
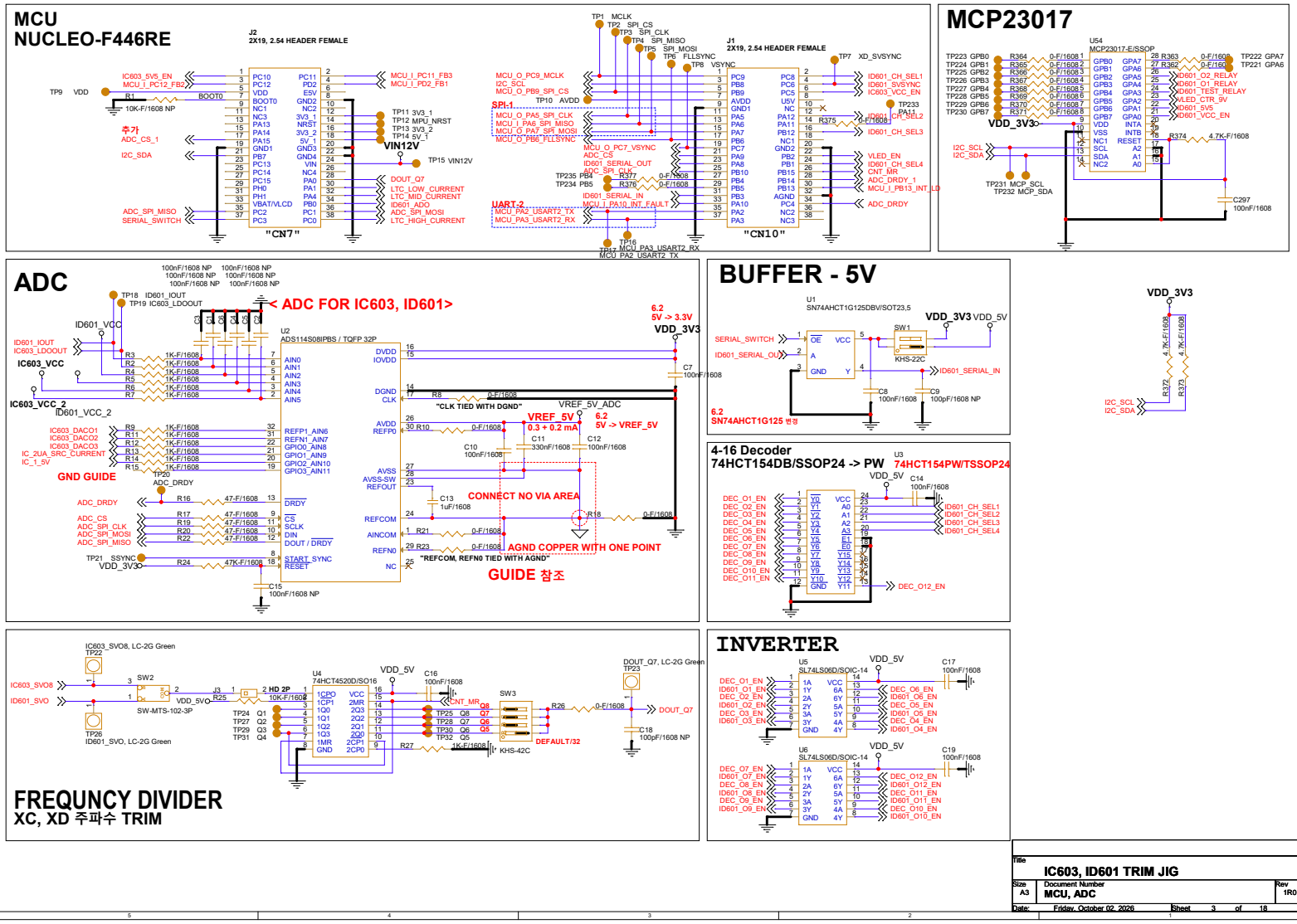
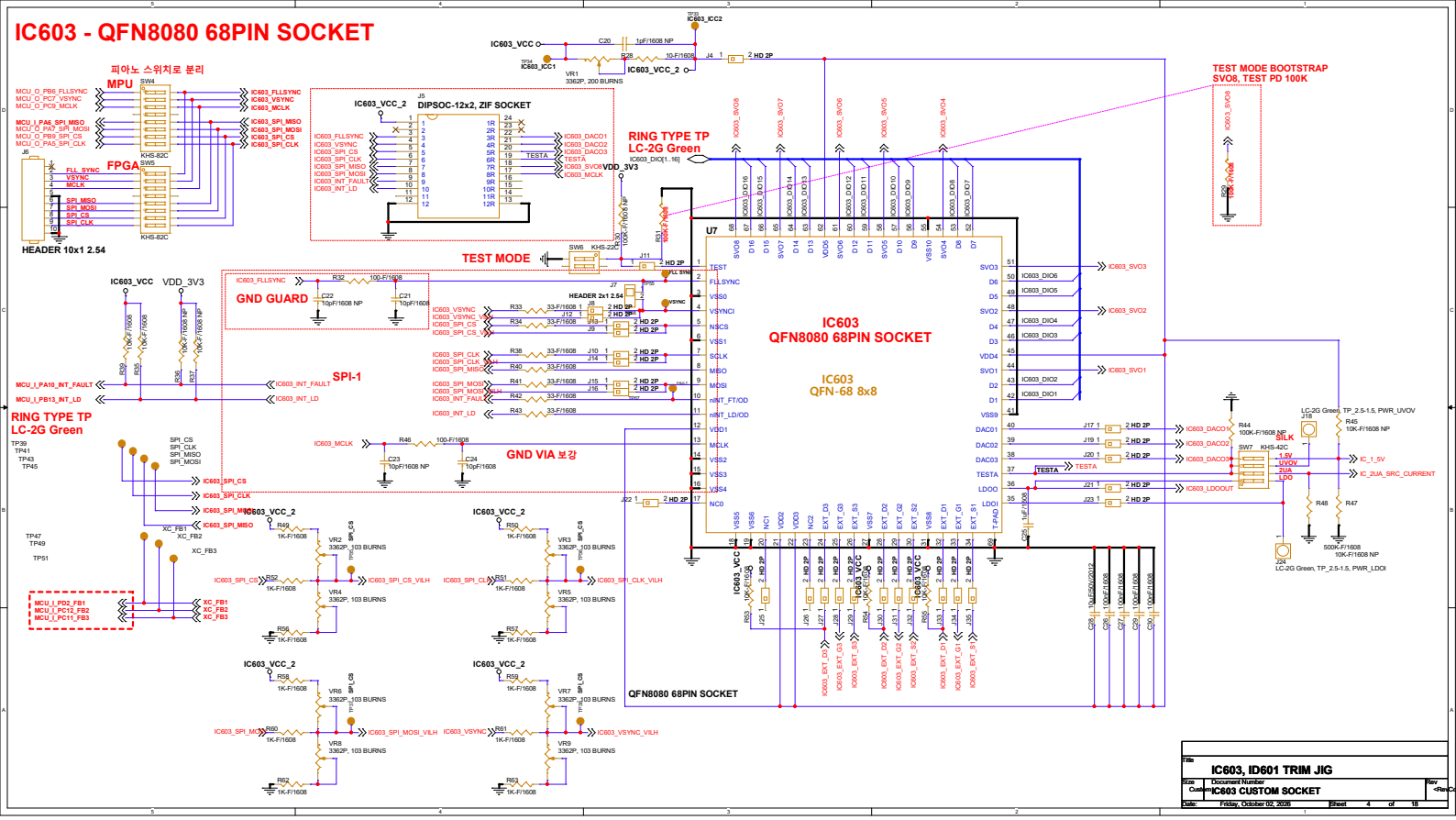
DATE	IC603 ID601 TRIM JIG
09.01	LOGIC DESIGN START
09.08	1차 수정안 IC603 68, 64PIN 적용 SVO8, 9, TEST : 100K PD 추가 DIOx, SVOx 33 -> pin header 2pin으로 변경 DIO_OUT, SV_OUT 부분에 33 추가 IC 주변 TP -> RING TYPE TP 0 -> 2P HEADER
09.08	2차 수정안 ID601 QFN24 3x3, TSSOP24 적용
09.10	3차 수정안 (REVISION No. 1R0) - XC24R/XD12R 삭제 P4,P9 - NET 변경 (XC -> IC603, XD-> ID601)
09.11	4차 수정안 - BLOCK UPDATE - IC603 QFN64 -> QFP64 - ID601 P.17 VSS6 -> TEST - XC24R_ADCO1~3 DAC 삭제 - SV08 주파수 분주 ; 32 - CURRENT SENSING ; MULTI-TURN RES. 수정
09.14	5차 수정안 - GATE / SOURCE 전압 측정 - ADC_DRDY_1, ADC_CS_1 추가
09.15	6차 수정안 - 09.14 DESIGN REVIEW 반영
09.17	7차 수정안 - 09.17 VIL / VIH MEASURE 반영
09.23	8차 수정안 IC603_ID601 TRIM JIG 1R0_8 - 09.23 : 1D0601 QFN24 40x40 추가

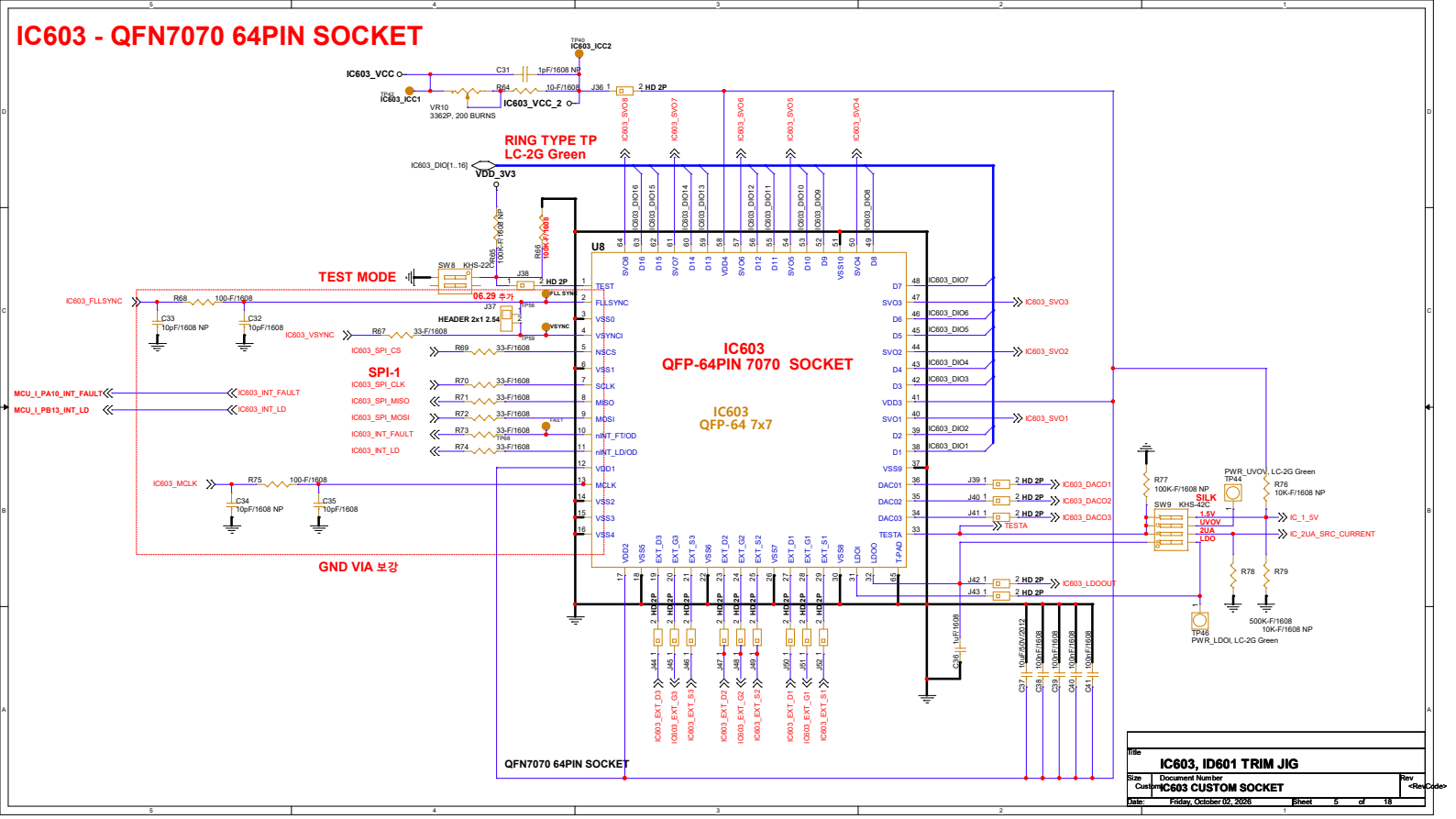
Title: IC603, ID601 TRIM JIG		
Size: Custom	Document Number: REVISION HISTORY	Rev: 1R0
Date: Friday, October 02, 2026	Sheet 1 of 18	

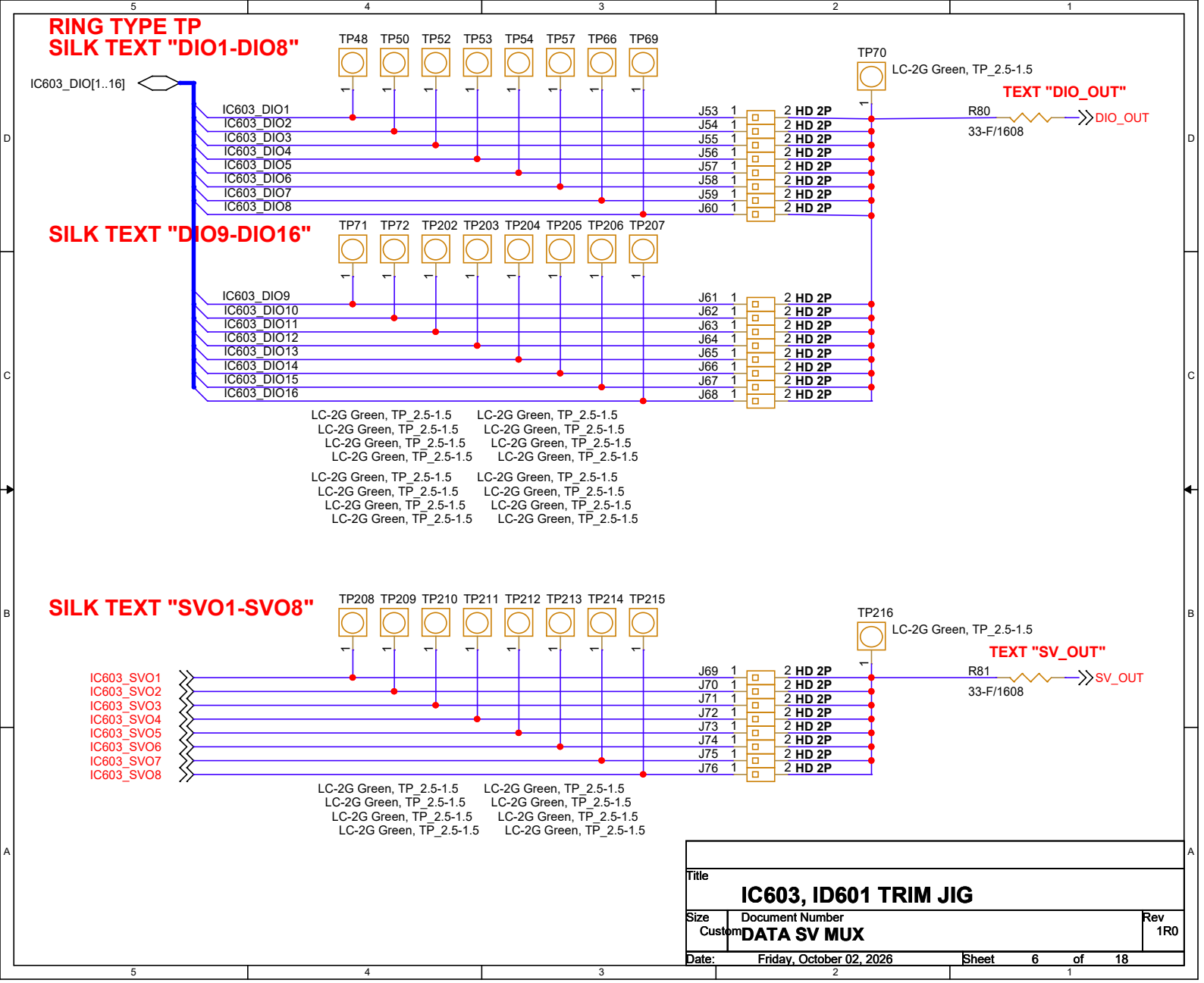

ID601+IC603
TRIM JIG 보드
PC
RS232
MCU board
(STM32F446RE NUCLEO)
전원: 12V
(3.3V내부연결)
MCU_DIO
MCU_SVI
MCU / IC603
SW
ID_DIO
ID_SVI
ID601#1
(TSSOP24)
7.8x6.4 mm
주문제작 Socket
Pin.
13
SVO
Frequency
Measure
O1
O12
Channel
Selection
Current
Measure
FPGA
Test 용
TP
FPGA / MCU
SW
SPI
MCLK
VSYNC
FLLSYNC
병렬연결
ID601#2 (QFN24)
3x3 mm
주문제작 Socket
ES 샘플 양시대응 用
ID601#3 (TSSOP24)
24Pin ZIF Socket
신뢰성 쪽 보드 용
DDIO
SVO
LED display (R,G,B chip)
Module
36 channels
IC603
Pin header Conn.
신뢰성 쪽보드 용
SPI
MCLK
VSYNC
FLLSYNC
병렬연결
IC603 (QFP 64)
7x7 mm
상용 Socket #1
D1 ··· D16
SVO1 ··· SVO8
D, SVO SELECT
SW
Dx
SVOx
ID601
#2
ZIF
Socket
DIO
SV
ID601
#3
ZIF
Socket
DIO
SV
ID601
#4
ZIF
Socket
DIO
SVO
HIGH_1,2,3 (GATE)
LOW_1,2,3 (GATE)
Power Switch Module
(시분할)
- Gate Driver
& FET
VLED
VLED R,G,B
Frequency
Measure
SVO ? (Pin.?)
Voltage
Measure
DACO1 (Pin.36)
DACO2 (Pin.35)
DACO3 (Pin.34)
TESTA (Pin. 1)
병렬연결
IC603 (QFN68)
8x8 mm
상용 Socket #2
Feed back
(DACO1, 2, 3)
RGB LED Power DC/DC,
Input : 12V,
VLED : 6V or 9V or 12V
Power Module
DC/DC
Input : 12V
Output : 5V, 3.3V, 1.8V,
1.5V
Control Switch &
LED Indication
IC603, ID601 TRIM JIG
BLOCK
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Title	IC603, ID601 TRIM JIG		
Size A3	Document Number MCU, ADC		Rev 1R0
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IC603 - QFN8080 68PIN SOCKET
피아노 스위치로 분리
MPU
FPGA
HEADER 10x1 2.54
IC603_VCC_2
DIPSOC-12x2, ZIF SOCKET
RING TYPE TP
LC-2G Green
TEST MODE
GND GUARD
SPI-1
GND VIA 보강
IC603
QFN-68 8x8
IC603
QFN8080 68PIN SOCKET
QFN8080 68PIN SOCKET
TEST MODE BOOTSTRAP
SVO8, TEST PD 100K
IC603, ID601 TRIM JIG
IC603 CUSTOM SOCKET
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IC603 - QFN7070 64PIN SOCKET
IC603_ICC2
IC603_VCC
C31
1pF/1608 NP
R64
10-F/1608
J36
2 HD 2P
IC603_ICC1
VR10
3362P, 200 BURNS
IC603_VCC_2
RING TYPE TP
LC-2G Green
IC603_DIO[1..16]
VDD_3V3
C665
100K-F/1608 NP
R66
100K-F/1608
U8
TEST MODE
SW8 KHS-22C
J38
2 HD 2P
FLL SYNC
06.29 추가
J37
HEADER 2x1 2.54
VSYNC
IC603_FLLSYNC
R68
100-F/1608
C33
10pF/1608 NP
C32
10pF/1608
IC603_VSYNC
R67
33-F/1608
IC603_SPI_CS
R69
33-F/1608
SPI-1
IC603_SPI_CLK
R70
33-F/1608
IC603_SPI_MISO
R71
33-F/1608
IC603_SPI_MOSI
R72
33-F/1608
IC603_INT_FAULT
R73
33-F/1608
IC603_INT_LD
R74
33-F/1608
MCU_I_PA10_INT_FAULT
MCU_I_PB13_INT_LD
IC603_INT_FAULT
IC603_INT_LD
IC603
QFP-64PIN 7070 SOCKET
IC603
QFP-64 7x7
IC603_MCLK
R75
100-F/1608
C34
10pF/1608 NP
C35
10pF/1608
GND VIA 보강
QFN7070 64PIN SOCKET
TEST
FLLSYNC
VSS0
VSYNCI
NSCS
VSS1
SCLK
MISO
MOSI
nINT_FT/OD
nINT_LD/OD
VDD1
MCLK
VSS2
VSS3
VSS4
VDD2
VSS5
EXT_D3
EXT_G3
EXT_S3
VSS6
EXT_D2
EXT_G2
EXT_S2
VSS7
EXT_D1
EXT_G1
EXT_S1
VSS8
LDOI
LDOO
T-PAD
D7
SVO3
D6
D5
SVO2
D4
D3
VDD3
SVO1
D2
D1
VSS9
DAC01
DAC02
DAC03
TESTA
IC603_DIO7
IC603_SVO3
IC603_DIO6
IC603_DIO5
IC603_SVO2
IC603_DIO4
IC603_DIO3
IC603_SVO1
IC603_DIO2
IC603_DIO1
J39
J40
J41
IC603_DACO1
IC603_DACO2
IC603_DACO3
TESTA
J42
J43
IC603_LDOOUT
R77
100K-F/1608 NP
SW9 KHS-42C
1.0V
UVOV
2UA
LDO
PWR_UVOV, LC-2G Green
TP44
R76
10K-F/1608 NP
IC_1_5V
IC_2UA_SRC_CURRENT
R78
R79
500K-F/1608
10K-F/1608 NP
TP46
PWR_LDOI, LC-2G Green
C38
1uF/1608
C37
10uF/50V/2012
C39
100nF/1608
C40
100nF/1608
C41
100nF/1608
IC603_EXT_D3
IC603_EXT_G3
IC603_EXT_S3
IC603_EXT_D2
IC603_EXT_G2
IC603_EXT_S2
IC603_EXT_D1
IC603_EXT_G1
IC603_EXT_S1
Title
IC603, ID601 TRIM JIG
Size
Custom
Document Number
IC603 CUSTOM SOCKET
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<RevCode>
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RING TYPE TP
SILK TEXT "DIO1-DIO8"
IC603_DIO[1..16]
TP48 TP50 TP52 TP53 TP54 TP57 TP66 TP69
IC603_DIO1
IC603_DIO2
IC603_DIO3
IC603_DIO4
IC603_DIO5
IC603_DIO6
IC603_DIO7
IC603_DIO8
J53 J54 J55 J56 J57 J58 J59 J60
2 HD 2P
TP70
LC-2G Green, TP_2.5-1.5
TEXT "DIO_OUT"
R80
33-F/1608
DIO_OUT
SILK TEXT "DIO9-DIO16"
TP71 TP72 TP202 TP203 TP204 TP205 TP206 TP207
IC603_DIO9
IC603_DIO10
IC603_DIO11
IC603_DIO12
IC603_DIO13
IC603_DIO14
IC603_DIO15
IC603_DIO16
J61 J62 J63 J64 J65 J66 J67 J68
LC-2G Green, TP_2.5-1.5
SILK TEXT "SVO1-SVO8"
TP208 TP209 TP210 TP211 TP212 TP213 TP214 TP215
IC603_SVO1
IC603_SVO2
IC603_SVO3
IC603_SVO4
IC603_SVO5
IC603_SVO6
IC603_SVO7
IC603_SVO8
J69 J70 J71 J72 J73 J74 J75 J76
TP216
TEXT "SV_OUT"
R81
SV_OUT
Title
IC603, ID601 TRIM JIG
Size
Custom
Document Number
DATA SV MUX
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Friday, October 02, 2026
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18

(Top View)

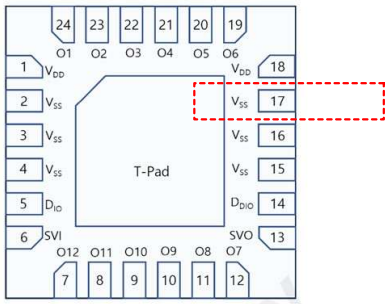
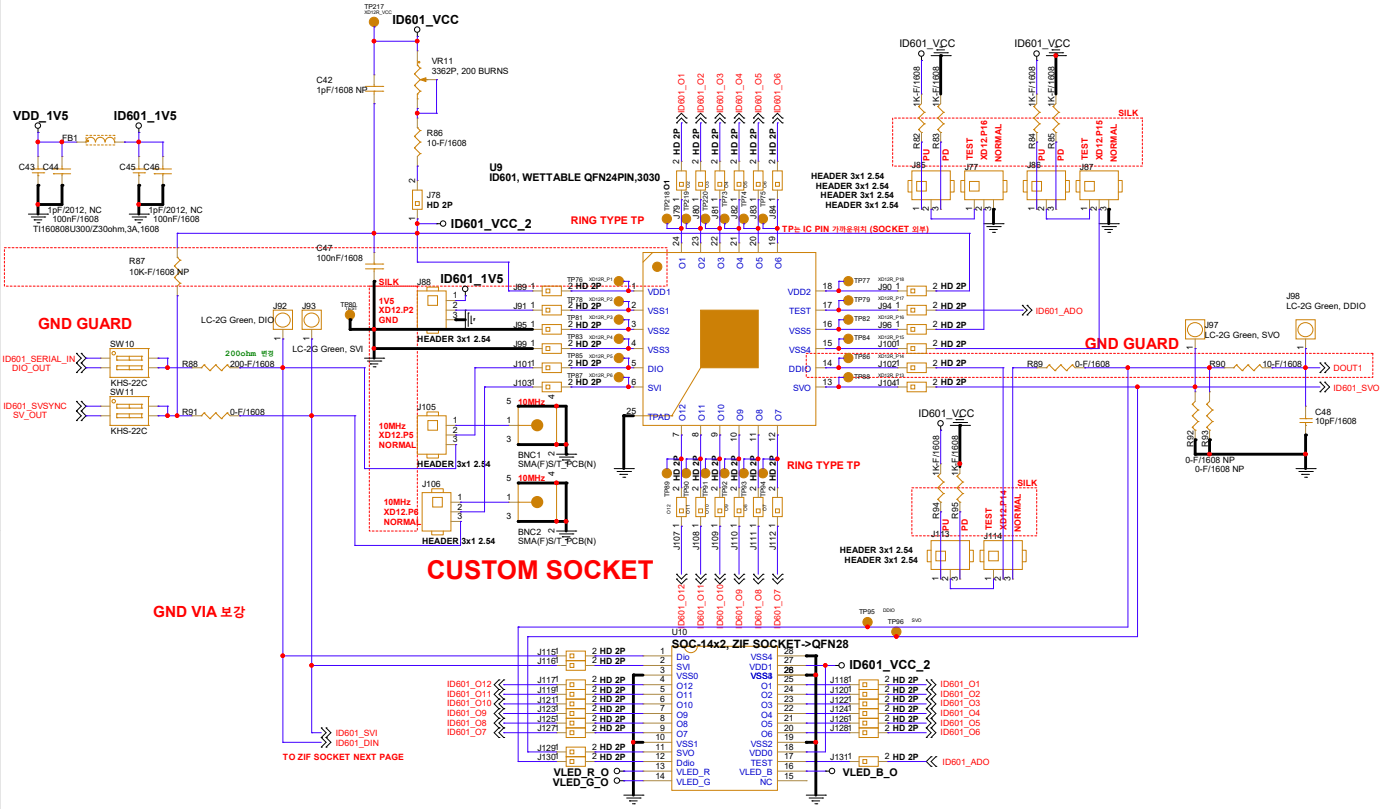


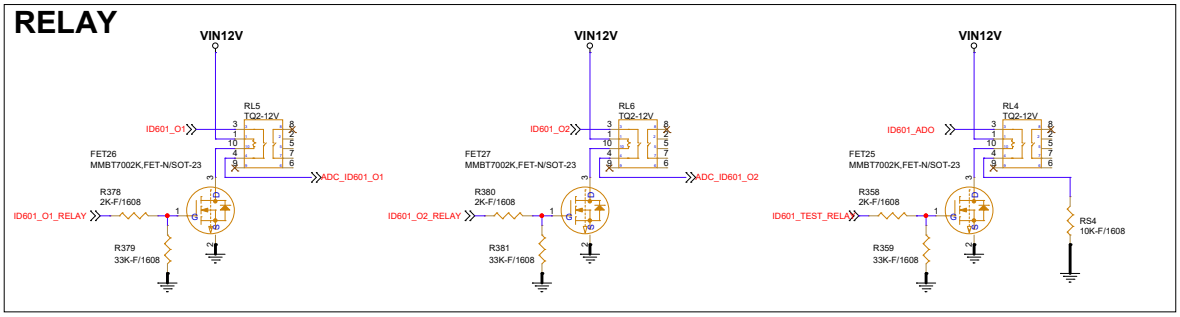
Figure 3. Pin diagram

Title	IC603, ID601 TRIM JIG	
Size: Custom	Document Number: ID601 PIN LAYOUT GUIDE	Rev: 1R0
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ID601 - QFN3030 24PIN SOCKET

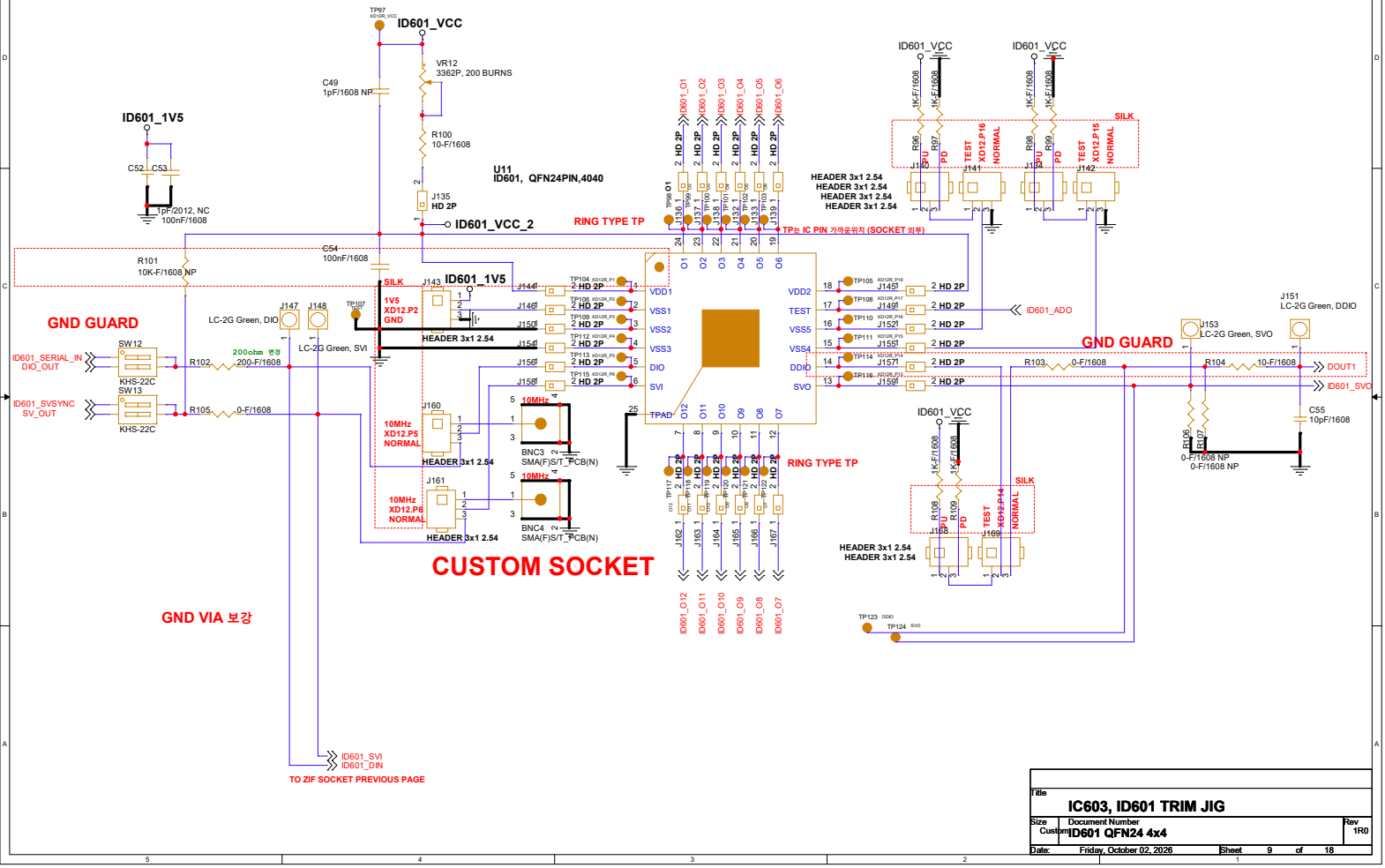


RELAY

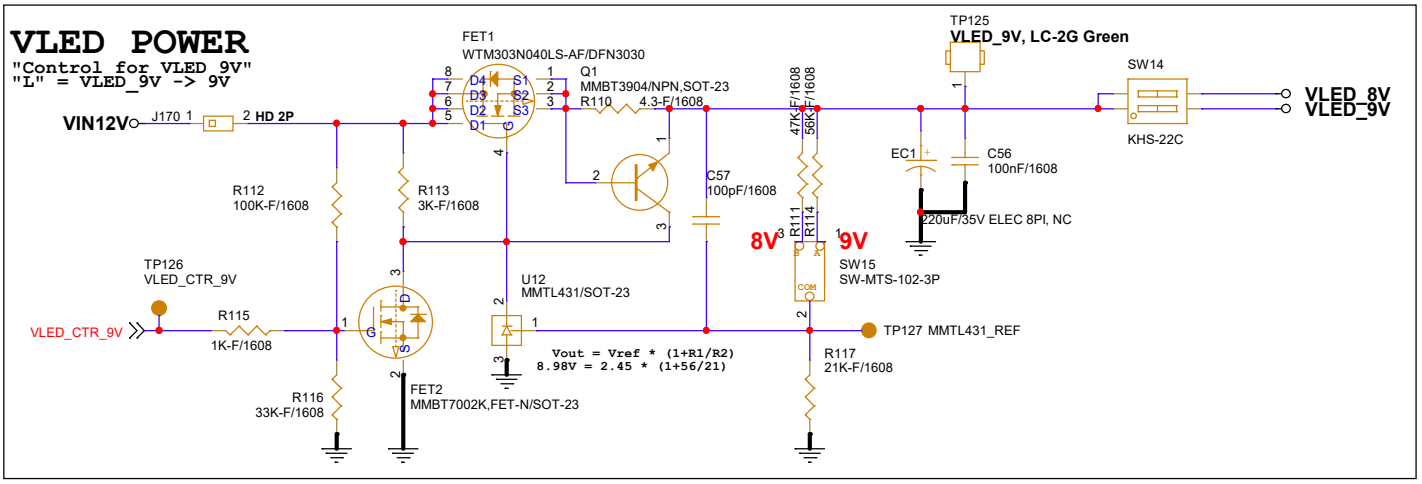


Title	IC603, ID601 TRIM JIG		
Size: Custom	Document Number: ID601 QFN24 3x3, ZIF (TSSOP24)		Rev: 1R0
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ID601 - QFN4040 24PIN SOCKET



Title	IC603, ID601 TRIM JIG		
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3PI DIP LED VF+1

RED Vf=2.0 If=20mA
GREEN Vf=3.0 If=20mA
BLUE Vf=2.5~3.2

VLED_R_O
VLED_B_O
VLED_G_O

3R4HT-7-4,3PI,R DAKWANG
3G4HT-7,3PI,G DAKWANG
3AB3HD00,3PI,B SMG

LED1 LED2 LED3 — KHS-42C SW16 — TP128 — ID601_O1
LED4 LED5 LED6 — KHS-42C SW17 — TP129 — ID601_O2
LED7 LED8 LED9 — KHS-42C SW18 — TP130 — ID601_O3
LED10 LED11 LED12 — KHS-42C SW19 — TP131 — ID601_O4
LED13 LED14 LED15 — KHS-42C SW20 — TP132 — ID601_O5
LED16 LED17 LED18 — KHS-42C SW21 — TP133 — ID601_O6

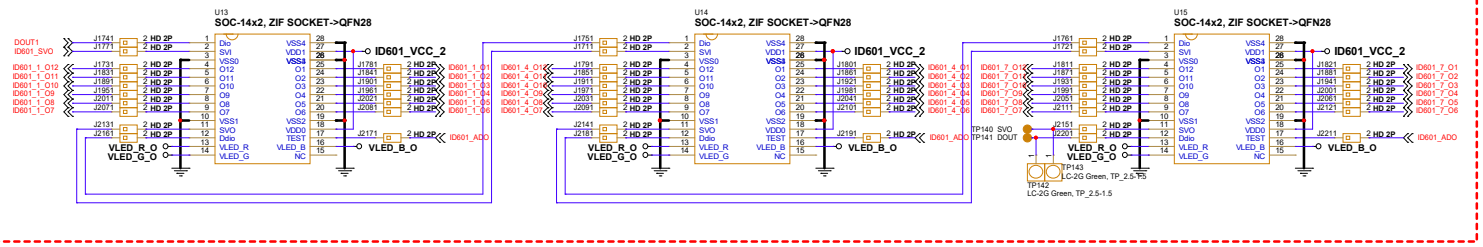
LC-2G Green, TP_2.5-1.5

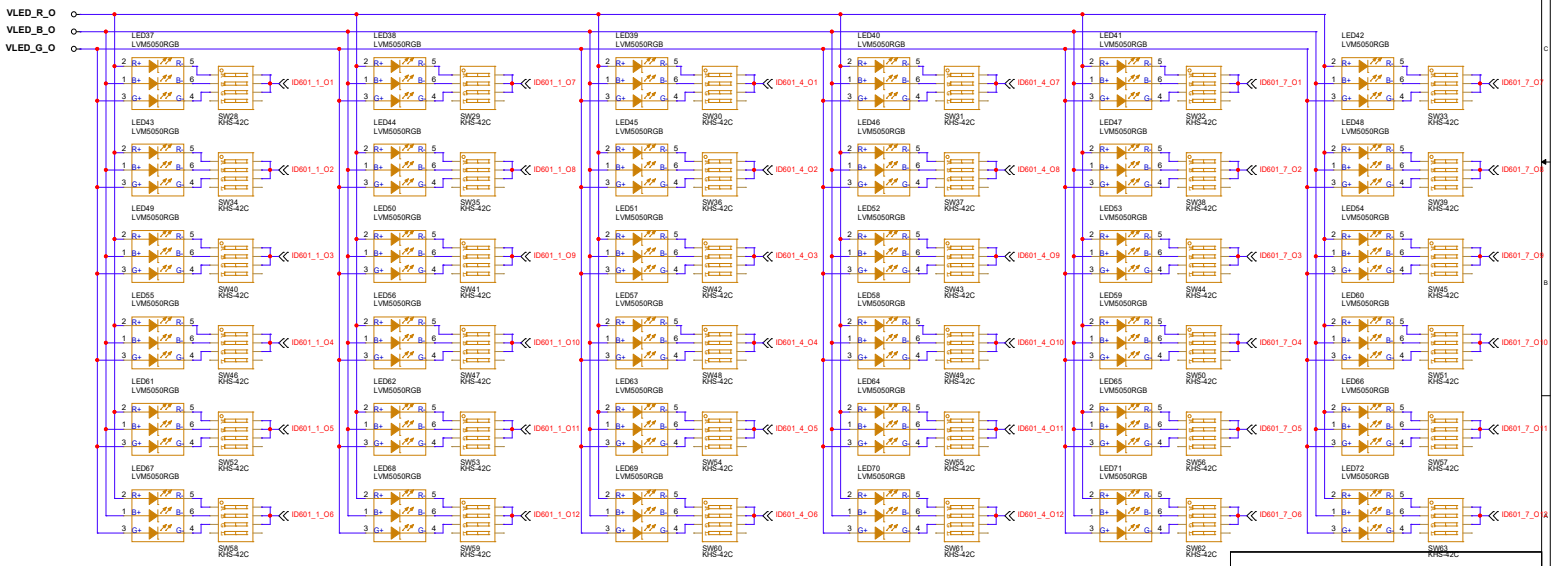
RG/B -> R/GB

LED19 LED20 LED21 — KHS-42C SW22 — TP134 — ID601_O7
LED22 LED23 LED24 — KHS-42C SW23 — TP135 — ID601_O8
LED25 LED26 LED27 — KHS-42C SW24 — TP136 — ID601_O9
LED28 LED29 LED30 — KHS-42C SW25 — TP137 — ID601_O10
LED31 LED32 LED33 — KHS-42C SW26 — TP138 — ID601_O11
LED34 LED35 LED36 — KHS-42C SW27 — TP139 — ID601_O12

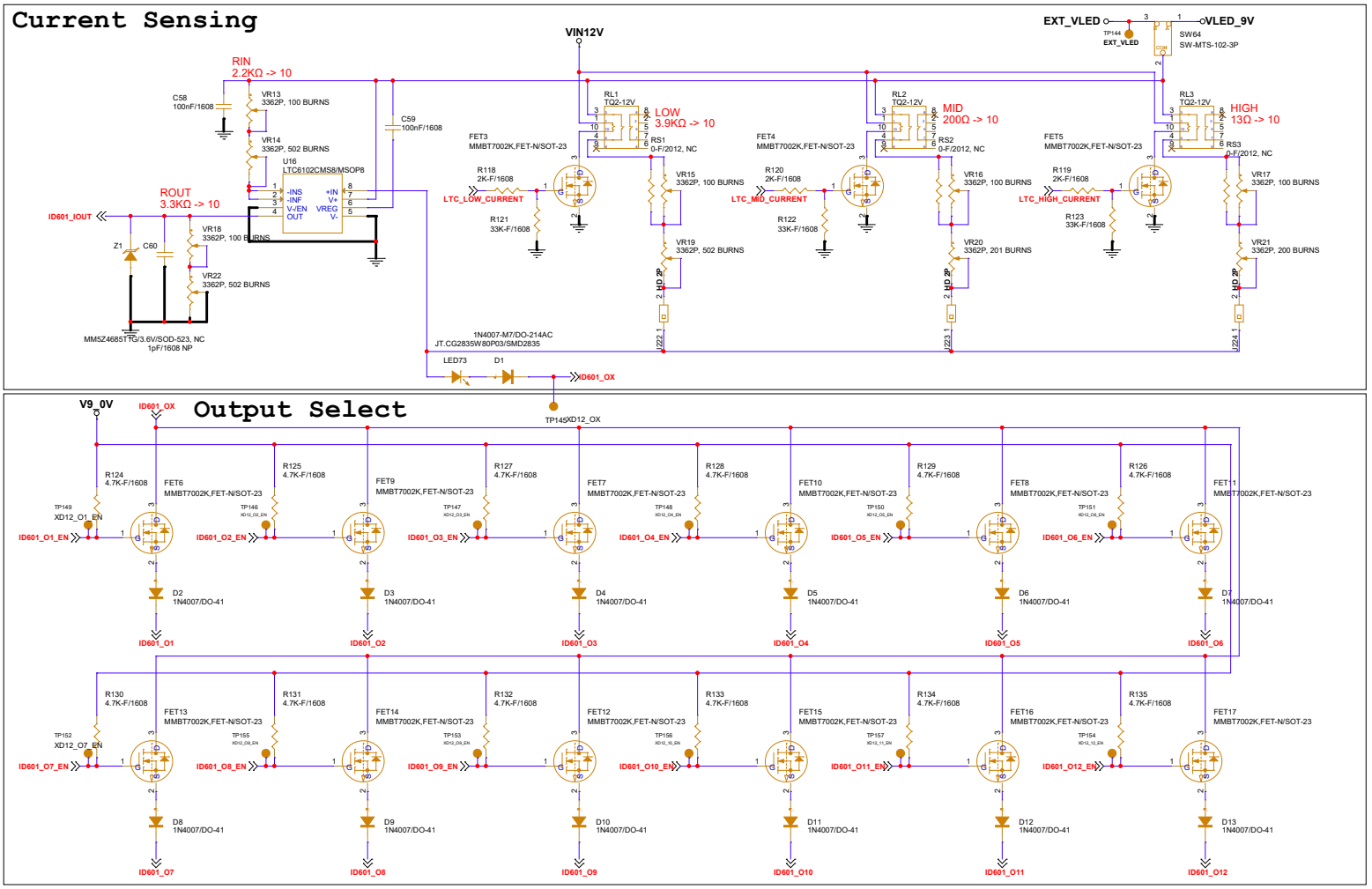
Title	IC603, ID601 TRIM JIG	
Size: Custom	Document Number: ID601 CASE BY TEST MODE SET	Rev: 1R0
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ID601 - ZIF SOCKET (ZIF SOCKET 간섭 고려)



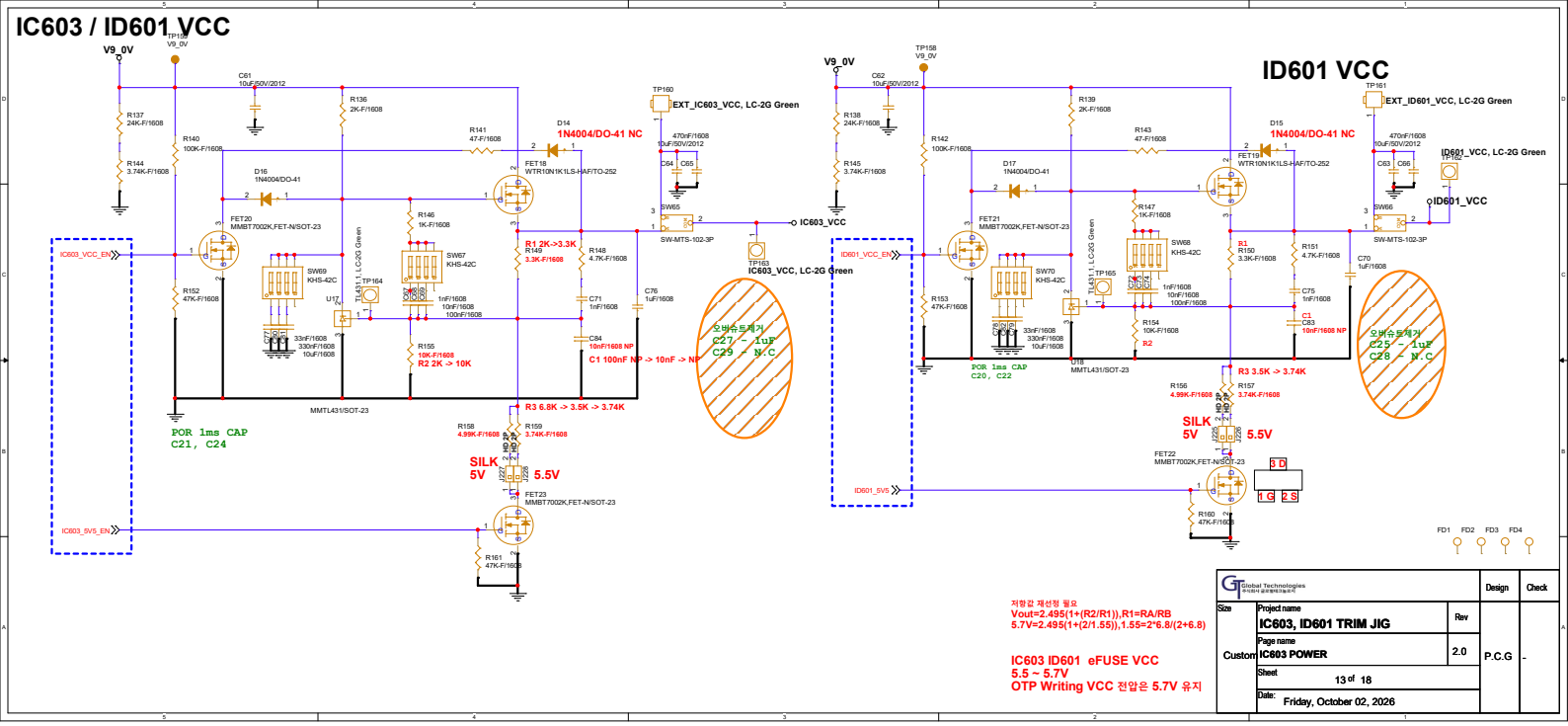


Title	IC603, ID601 TRIM JIG		
Size: Custom	Document Number: ID601 RGB LED		Rev: 1R0
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Global Technologies 주식회사 글로벌테크놀로지			Design	Check
Size: Custom	Project name: IC603, ID601 TRIM JIG	Rev: 2.0	P.C.G	-
	Page name: CURRENT MEASURE			
	Sheet: 12 of 18			
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IC603 / ID601 VCC



ID601 VCC

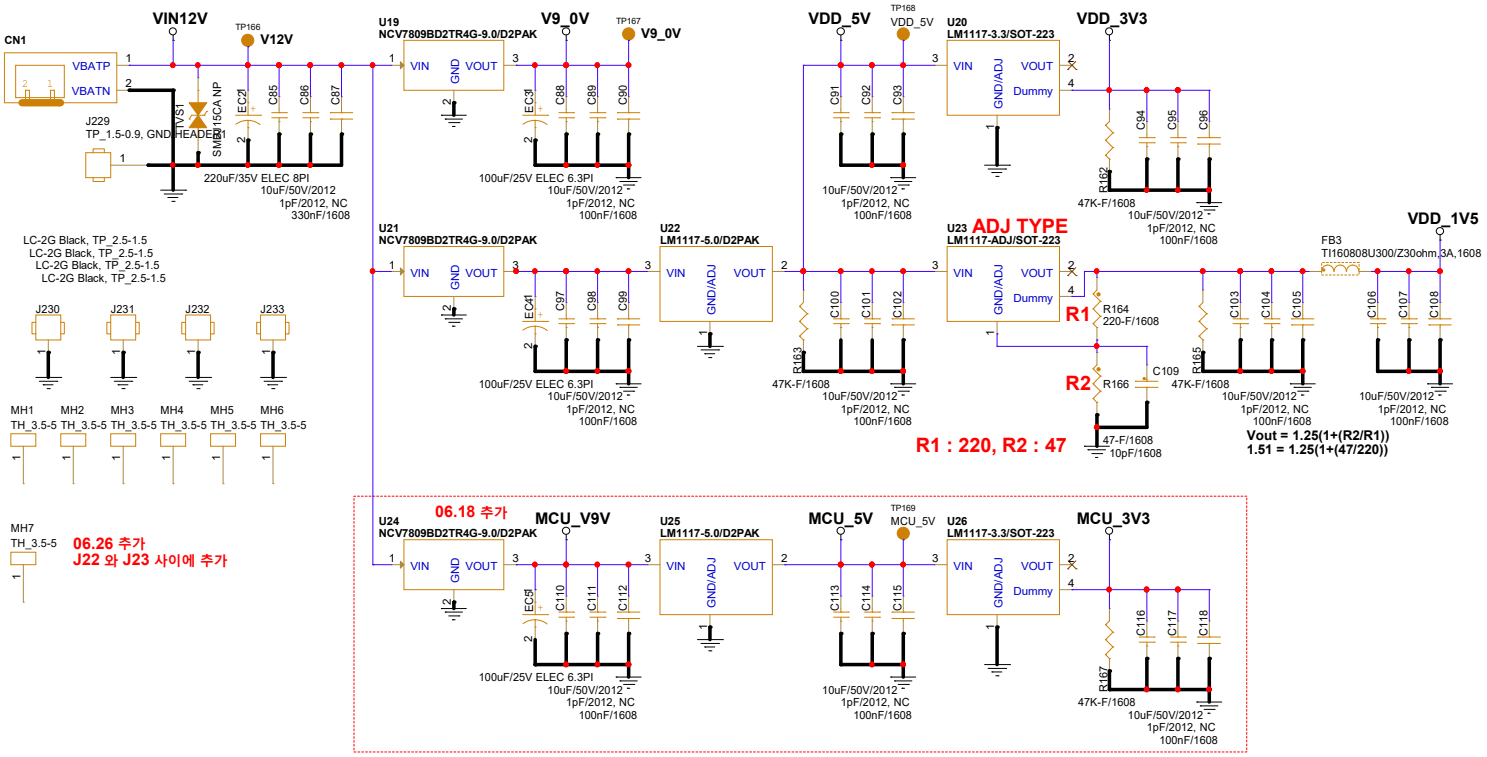
저항값 재선정 필요
Vout=2.495(1+(R2/R1)),R1=RA/RB
5.7V=2.495(1+(2/1.55)),1.55=2*6.8/(2+6.8)

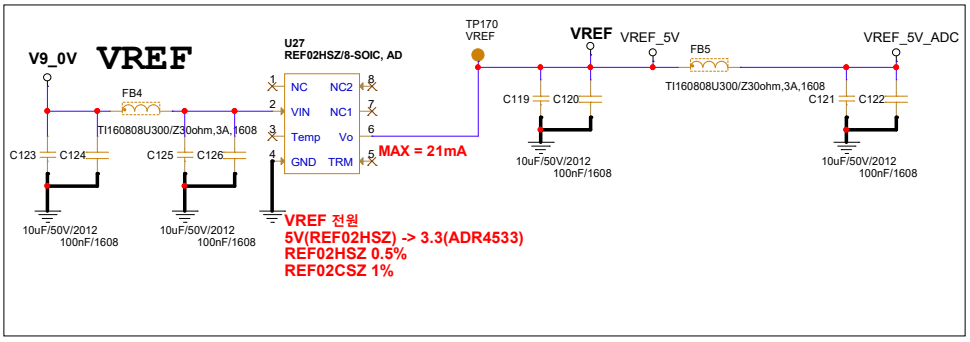
IC603 ID601 eFUSE VCC
5.5 ~ 5.7V
OTP Writing VCC 전압은 5.7V 유지

Global Technologies			Design	Check
Size: Custom	Project name: IC603, ID601 TRIM JIG	Rev: 2.0	P.C.G	-
	Page name: IC603 POWER			
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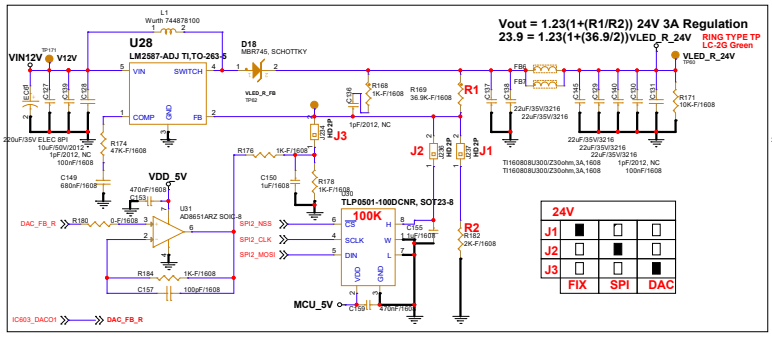
LOGIC POWER STANDBY

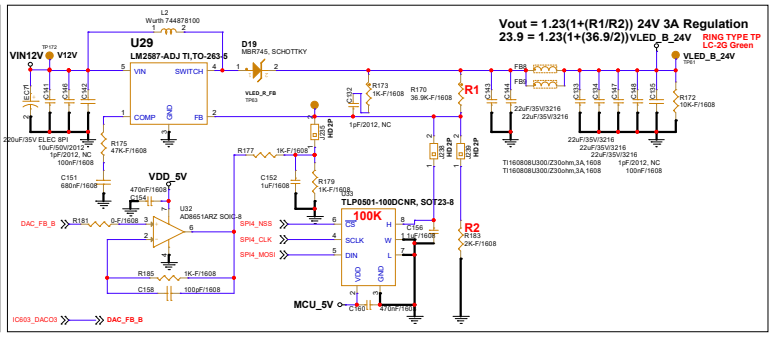
YW396-02V/1R*2-2pin,3.96mm,ST,#16~24,7.96A,DIP,Mate YH396-02V

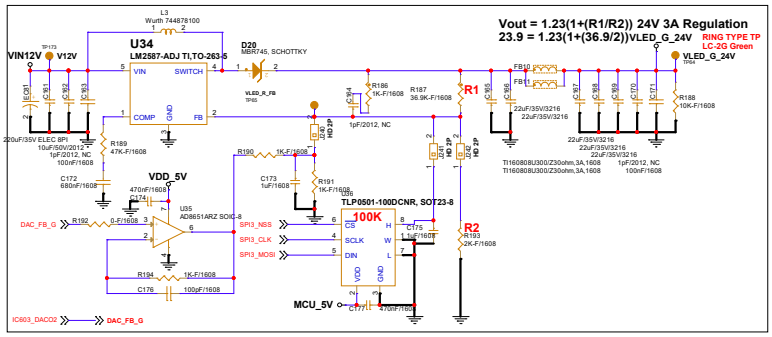




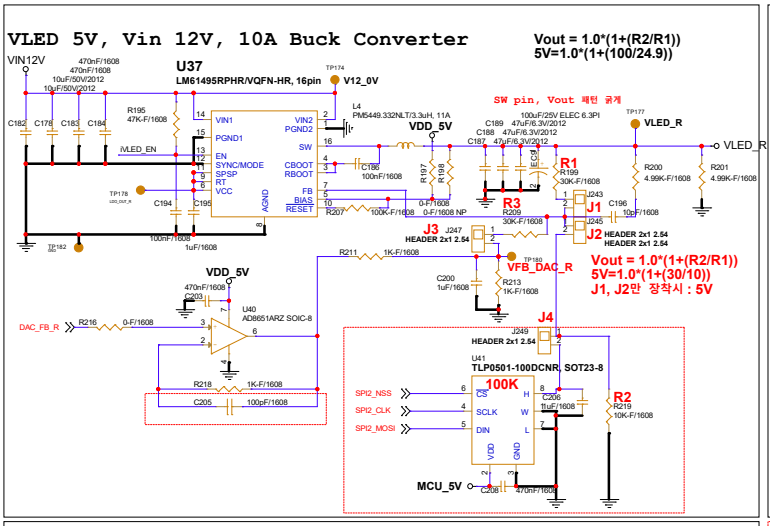
Global Technologies 주식회사 글로벌테크놀로지		Design	Check
Size: Custom	Project name: IC601, ID603 TRIM JIG	Rev: 1R0	P.C.G / -
	Page name: LOGIC POWER		
	Sheet: 14 of 18		
	Date: Friday, October 02, 2026		

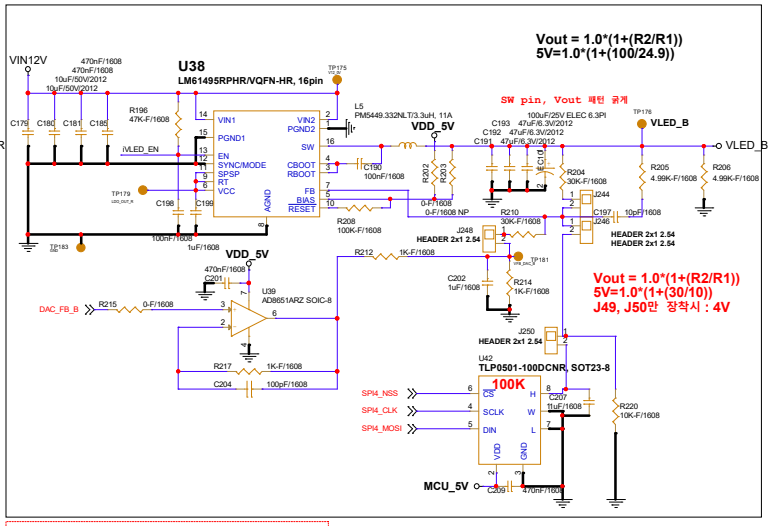


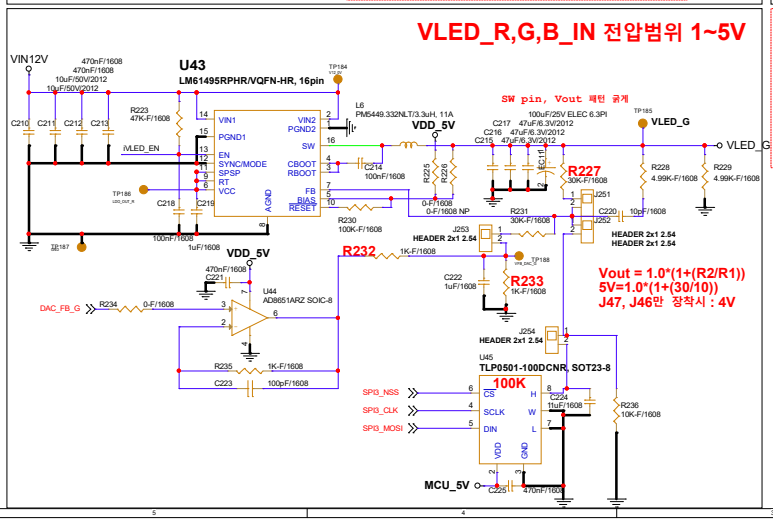


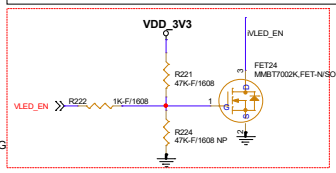


Global Technologies (India) pvt ltd		Design	Check
Size: Custom	Project name: IC601, ID603 TRIM JIG	Rev: 1R0	P.C.G
	Page name: VLED 24V POWER		-
	Sheet: 15 of 18		
	Date: Friday, October 02, 2026		

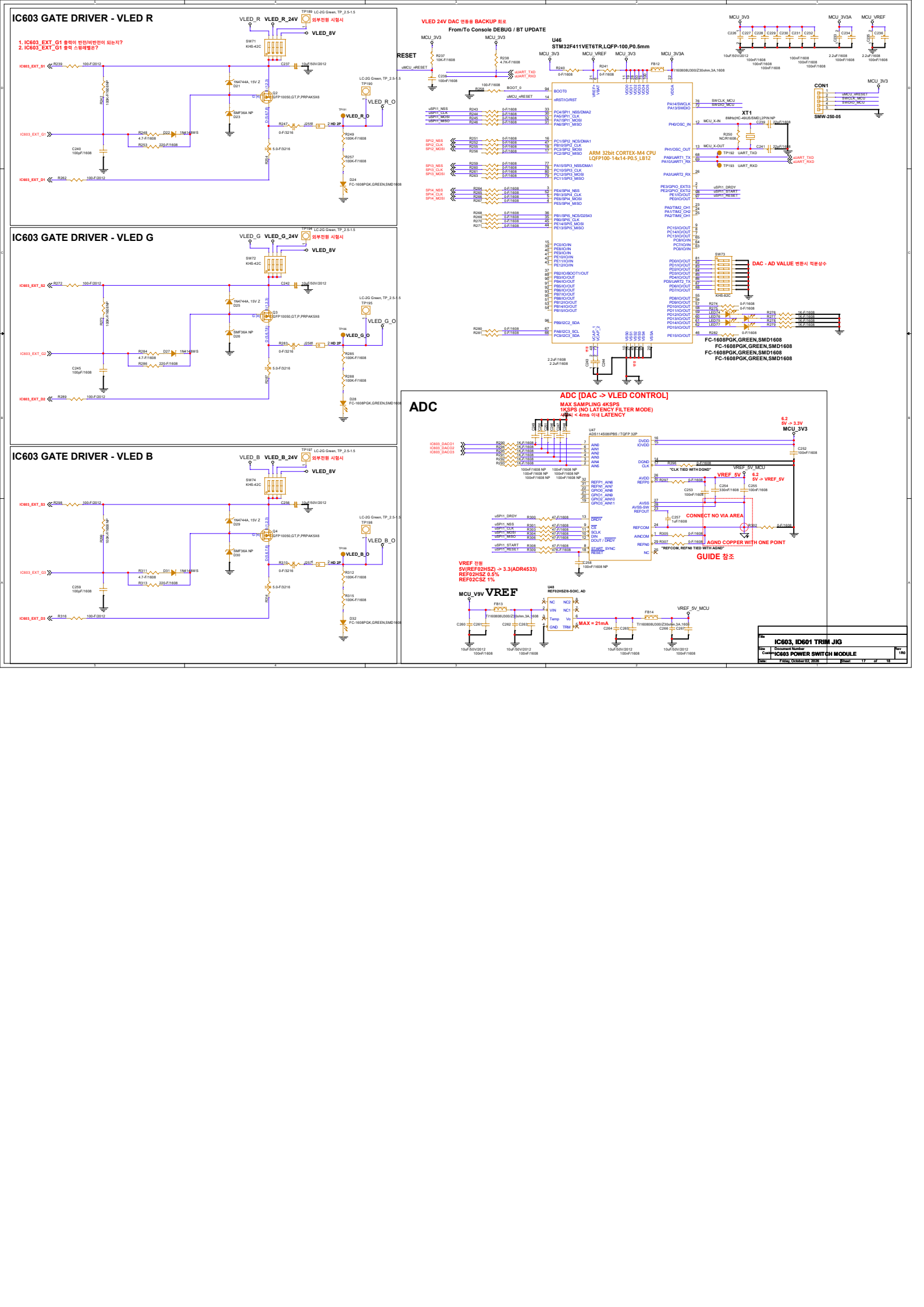


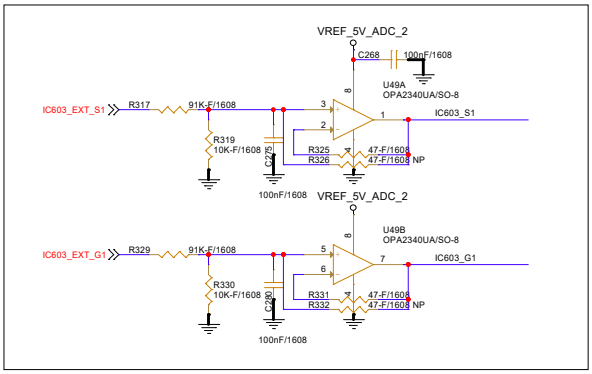


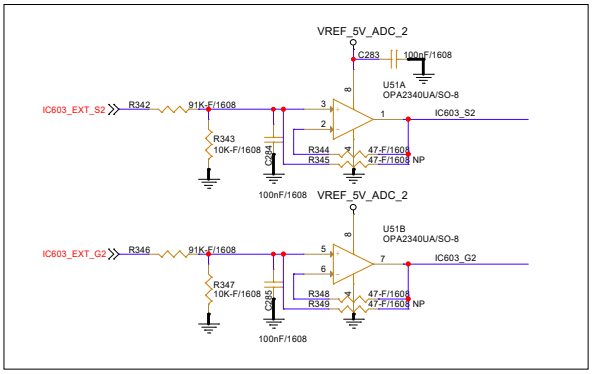


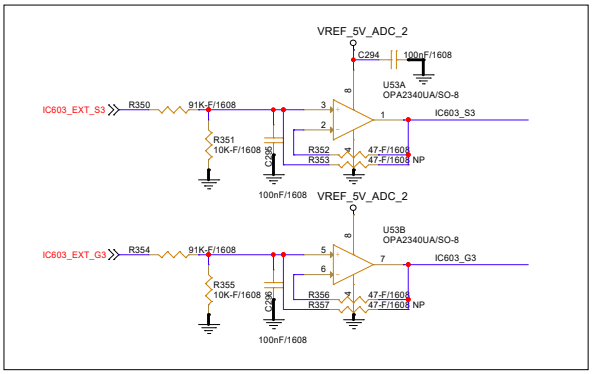


Title	IC603, ID601 TRIM JIG		
Size: Custom	Document Number: VLED 5V POWER		Rev: 1R0
Date: Friday, October 02, 2026	Sheet 16 of 18		


IC603 GATE DRIVER - VLED R
VLED_R
VLED_R_24V
외부전원 시험시
VLED_8V
1. IC603_EXT_G1 출력이 반전/비반전이 되는지?
2. IC603_EXT_G1 출력 스윙레벨은?
IC603_EXT_S1
IC603_EXT_G1
IC603_EXT_D1
VLED_R_O
IC603 GATE DRIVER - VLED G
VLED_G
VLED_G_24V
외부전원 시험시
VLED_8V
IC603_EXT_S2
IC603_EXT_G2
IC603_EXT_D2
VLED_G_O
IC603 GATE DRIVER - VLED B
VLED_B
VLED_B_24V
외부전원 시험시
VLED_8V
IC603_EXT_S3
IC603_EXT_G3
IC603_EXT_D3
VLED_B_O
VLED 24V DAC 연동용 BACKUP 회로
From/To Console DEBUG / BT UPDATE
RESET
U46
STM32F411VET6TR,LQFP-100,P0.5mm
ARM 32bit CORTEX-M4 CPU
LQFP100-14x14-P0.5_LB12
XT1
8MHz(HC-49US/SMD) 12PIN NP
CON1
SMW-250-05
DAC - AD VALUE 변환시 적분상수
FC-1608PGK,GREEN,SMD1608
ADC
ADC [DAC -> VLED CONTROL]
MAX SAMPLING 4KSPS
1KSPS (NO LATENCY FILTER MODE)
ADC 입력 < 4ms 이내 LATENCY
U47
ADS114S08IPBS / TQFP 32P
"CLK TIED WITH DGND"
CONNECT NO VIA AREA
AGND COPPER WITH ONE POINT
"REFCOM, REFN0 TIED WITH AGND"
GUIDE 참조
VREF 전원
5V(REF02HSZ) -> 3.3(ADR4533)
REF02HSZ 0.5%
REF02CSZ 1%
U48
REF02HSZI8-SOIC, AD
MAX = 21mA
IC603, ID601 TRIM JIG
IC603 POWER SWITCH MODULE
Friday, October 02, 2026
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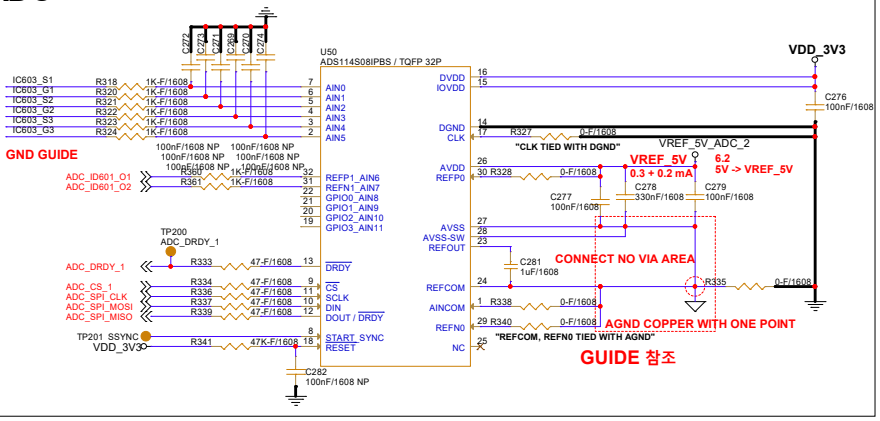






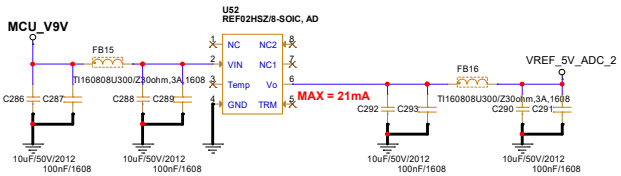
ADC

< ADC FOR IC603, SOURCE / GATE MEASURE >



VREF

VREF 전원
5V(REF02HSZ) -> 3.3(ADR4533)
REF02HSZ 0.5%
REF02CSZ 1%



Title	IC603, ID601 TRIM JIG	
Size: Custom	Document Number: IC603 GATE SOURCE MEASURE	Rev: 1R0
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